

LECTURE 1 (SEPTEMBER 1ST)

Definition. (Density operator) A general quantum state is represented by a density operator.

$$\hat{\rho} = \sum_{i=1}^n p_i |\psi_i\rangle \langle \psi_i|$$

where $\hat{\rho}$ is

- (i) Hermitian $\hat{\rho} = \hat{\rho}^\dagger$
- (ii) Positive operator $\langle \psi | \hat{\rho} | \psi \rangle \geq 0$ for an arbitrary $|\psi\rangle$
- (iii) Trace is 1 $\text{Tr}(\hat{\rho}) = 1$

It is easy to think the operator that gives classical probability weighting when a measurement is preformed. For example, take an experiment that either has a spin up or spin down as its output. Generally, an output would be expressed as a superposition,

$$P_\uparrow \langle z; \uparrow | \hat{O} | z; \uparrow \rangle + P_\downarrow \langle z; \downarrow | \hat{O} | z; \downarrow \rangle$$

where $P_\uparrow$ and $P_\downarrow$. Notice that by taking the above operator as the definition of a mixed state, we can simply take

$$\text{Tr}(\hat{\rho} \hat{O})$$

to obtain the expected value of a certain observable. The thing is, there are two types of uncertainty – classical and quantum.

Definition. (Quantum von Neumann entropy) Classically, we define the Shannon entropy for a probability distribution $\{p_i\}$ as

$$H(p) = - \sum_i p_i \ln p_i$$

Similarly, we define the von Neumann entropy has

$$S(\rho) = - \text{Tr}(\rho \ln \rho)$$

Notice that when ρ has eigenvalues $\{\lambda_i\}$,

$$S(\rho) = - \sum_i \lambda_i \ln \lambda_i$$

Remark. (Pure states and equivalences *) Notice that a pure state can be expressed as a density operator like the following

$$\hat{\rho} = |\psi\rangle \langle \psi|$$

These types of operators are idempotent ($\rho^2 = \rho$) meaning that their expected value is

either 1 or 0. Now notice that two density matrices of pure states are equivalent, that is,

$$|x\rangle\langle x| = |y\rangle\langle y|$$

if and only if

$$|x\rangle = |y\rangle\langle y|x\rangle$$

where

$$1 = |\langle x|y\rangle|^2$$

and the states differ purely by some phase factor $\exp(i\theta)$ as we would expect.

LECTURE 2 (SEPTEMBER 3RD)

Recall. We have learned that the expectation value of a certain observable was given by

$$\text{Tr}(\hat{\rho}\hat{A})$$

Using the spectral decomposition $\hat{A} = \sum a|a\rangle\langle a| = \sum a \text{Pr}_a$, we can express the probability of observing $\hat{A}$ and obtaining a as

$$\text{Tr}(\hat{\rho} \text{Pr}_a)$$

You should check how this works for a pure state? Notice that

$$\frac{\text{Pr}_a \hat{\rho} \text{Pr}_a}{\text{Tr}(\hat{\rho} \text{Pr}_a)} = \frac{1}{\text{Tr}(\hat{\rho} \text{Pr}_a)} \text{Tr}(\text{Pr}_a \hat{\rho} \text{Pr}_a) = 1$$

and that the trace is 1.

Example. We check that the density matrix satisfies

$$\begin{aligned} \dot{\rho} &= \frac{d}{dt}(|\psi\rangle\langle\psi|) \\ &= |\dot{\psi}\rangle\langle\psi| + |\psi\rangle\langle\dot{\psi}| \\ &= -\frac{i}{\hbar}H|\psi\rangle\langle\psi| + \frac{i}{\hbar}|\psi\rangle\langle\psi|H \\ &= -\frac{i}{\hbar}[H, \rho] \end{aligned}$$

Remark. (General properties of $\hat{\rho}$) Notice that the square of the density matrix is equal to

$$\hat{\rho}^2 = \sum_j p_j^2 |\psi_j\rangle\langle\psi_j|$$

Therefore, in general, we say mixed states are states whose density matrix satisfies

$$\hat{\rho} \neq \hat{\rho}^2$$

and a pure state otherwise. Also,

$$\hat{\rho} - \hat{\rho}^2 = \sum_j (p_j - p_j^2) |\psi_j\rangle\langle\psi_j|$$

and the expected value is always bigger than 0. Sometimes, we write $\hat{\rho} \geq \hat{\rho}^2$. We also ask, what is the norm of $|\hat{\rho}\psi\rangle$ for an arbitrary $|\psi\rangle$? We notice that

$$\langle\hat{\rho}\psi|\hat{\rho}\psi\rangle = \langle\psi|\hat{\rho}^2|\psi\rangle = \langle\psi|\rho\psi\rangle \leq \sqrt{\langle\psi|\psi\rangle} \sqrt{\langle\hat{\rho}\psi|\hat{\rho}\psi\rangle}$$

Resulting in

$$\langle\hat{\rho}\psi|\hat{\rho}\psi\rangle \leq \langle\psi|\psi\rangle$$

We see that the density matrix is a contraction mapping. We now prove the contraposition – if $\hat{\rho}^2 = \hat{\rho}$, the state is pure. Last class we have seen that the eigenvalue of the matrix becomes either 1 or 0. In the spectral decomposition,

$$\hat{\rho} = \sum_j p_j |\psi_j\rangle\langle\psi_j|$$

we see that only one of the p_j 's should become one and that the others should be zero. We have now concluded the proof.

Theorem. (Relationship between diagonal and non-diagonal density matrices) Always, you want to find the diagonal form of an operator. How do you find it from a mixed state?

$$\hat{\rho} = \sum_{i=1}^n p_i |\psi_i\rangle\langle\psi_i| \longleftrightarrow \hat{\rho} = \sum_{j=1}^m w_j |\alpha_j\rangle\langle\alpha_j|$$

We obviously know that $n \leq m$. Take

$$\langle\psi_k|\hat{\rho}|\psi_k\rangle = 0$$

Then,

$$\sum_{j=1}^N w_j |\langle\psi_i|\alpha_j\rangle|^2 = 0$$

which tells us that

$$\langle\psi_i|\alpha_j\rangle = 0$$

Therefore,

$$|\alpha_j\rangle = \sum_{k=1}^n \langle\psi_k|\alpha_j\rangle |\psi_k\rangle$$

We now define a $m \times n$ matrix

$$V_{ij} = \sqrt{\frac{w_i}{p_j}} \langle\psi_j|\alpha_i\rangle$$

It is obvious that

$$\sum_{j=1}^n V_{ij} \sqrt{p_j} |\psi_j\rangle = \sqrt{w_i} |\alpha_i\rangle$$

and we have

$$\sum_{i=1}^m (V^\dagger)_{ji} V_{ij} = \frac{1}{\sqrt{p_j p_k}} \sum_{i=1}^m w_i \langle \psi_j | \alpha_i \rangle \langle \alpha_i | \psi_k \rangle = \frac{1}{\sqrt{p_j p_k}} \langle \psi_j | \rho^2 | \psi_k \rangle = \delta_{jk}$$

LECTURE 3 (SEPTEMBER 8TH)

Example. Consider the Hilbert space $\mathcal{H} = \mathcal{C}^2$ of a qubit system. An arbitrary mixed state is expressed as

$$\rho = \sum_{i=1}^2 w_i |\phi_i\rangle \langle \phi_i|$$

Consider

$$\begin{cases} |\phi_1\rangle = |x; \uparrow\rangle = \frac{1}{2} \begin{pmatrix} 1 \\ 1 \end{pmatrix} \\ |\phi_2\rangle = |z; \uparrow\rangle = \begin{pmatrix} 1 \\ 0 \end{pmatrix} \end{cases}$$

with $w_1 = 2/5$ and $w_2 = 3/5$. The density matrix is then given as

$$\begin{aligned} \rho &= \frac{2}{5} \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ 1 \end{pmatrix} \frac{1}{\sqrt{2}} (1 \ 1) + \frac{3}{5} \begin{pmatrix} 1 \\ 0 \end{pmatrix} (1 \ 0) \\ &= \frac{1}{5} \begin{pmatrix} 4 & 1 \\ 1 & 1 \end{pmatrix} \end{aligned}$$

Calculating ρ^2 we find that $\rho^2 \neq \rho$ and that ρ is a mixed state. It is a positive operator as

$$\rho - \rho^2 = \frac{1}{25} \begin{pmatrix} 3 & 0 \\ 0 & 3 \end{pmatrix}$$

We can lastly perform a spectral decomposition with $p_{1,2} = 1/2 \pm \sqrt{13}/10$ and

$$\phi = p_1 |1\rangle \langle 1| + p_2 |2\rangle \langle 2| = \rho$$

Other examples are cooling and optical.

Definition. (Perturbation theory) There are very few quantum systems that we can solve analytically. Perturbation theory provides a way to approximate quantum results and is very important. Let's first look at the big picture. An arbitrary Hamiltonian can be written

as

$$\hat{H}(t) = \hat{H}_0 + \lambda \hat{V}(t)$$

where we fully know $\hat{H}_0$ as in the eigenvalues, eigenkets and etcetera. We should also take λ to be very small for this to make sense. There are large two classes, (1) time independent perturbations and (2) time dependent perturbations. Each are defined as perturbations where $\hat{V}$ is independent and dependent of time. Let's deal with time independent perturbation theory for now.

Example. (Non-degenerate perturbation) Given the information of $\hat{H}_0$ we want to inquire the following,

$$E_n^{(0)} \quad \& \quad |n^{(0)}\rangle \quad \implies \quad |n\rangle_\lambda \quad \& \quad E_n$$

where $\hat{H}|n\rangle_\lambda = E_n|n\rangle_\lambda$. We automatically expect

$$E_n = E_n^{(0)} + \Delta_n(\lambda)$$

and thus

$$(\hat{H}_0 - E_n^{(0)})|n\rangle_\lambda = (\Delta_n(\lambda) - \lambda \hat{V})|n\rangle_\lambda$$

As $H_0|n^{(0)}\rangle = E_n^{(0)}|n^{(0)}\rangle$, $\langle n^{(0)}|(H_0 - E_n^{(0)}) = 0$ which leads to

$$\langle n^{(0)}|\Delta_n(\lambda) - \lambda \hat{V}|n\rangle_\lambda = 0$$

Telling us that $|n^{(0)}\rangle$ is missing from the ket of the right-hand side. Now define the projection from

$$\mathbf{1} = \sum_k |k^{(0)}\rangle \langle k^{(0)}|$$

where the Hilbert space $\mathcal{H}$ has the basis $\{|k^{(0)}\rangle\}$ as

$$\phi_n = \mathbf{1} - |n^{(0)}\rangle \langle n^{(0)}| = \sum_{k \neq n} |k^{(0)}\rangle \langle k^{(0)}|$$

gives some n . Now,

$$|n\rangle_\lambda = C_n(\lambda)|n^{(0)}\rangle + \frac{1}{E_n^{(0)} - \hat{H}_0} \phi_n (-\Delta_n(\lambda) + \lambda \hat{V})|n\rangle_\lambda$$

Notice that $\langle n^{(0)}|n\rangle_\lambda = C_n(\lambda)$ and that $C_n(\lambda)$ dictates normalisation (how much $|n^{(0)}\rangle$ is in $|n\rangle_\lambda$). We expect the following

$$|n\rangle_\lambda = C_n(\lambda) \left(|n^{(0)}\rangle + \lambda |n^{(1)}\rangle + \lambda^2 |n^{(2)}\rangle + \dots \right)$$

now we of course want to substitute.

LECTURE 4 (SEPTEMBER 10TH)

Recall. The perturbed ket satisfied the relation

$$(\lambda \hat{V} - \Delta_n(\lambda))|n\rangle_\lambda = (E_n^{(0)} - \hat{H}_0)|n\rangle_\lambda$$

Applying $\langle n^{(0)}|$,

$$\langle n^{(0)}|\lambda \hat{V}|n\rangle_\lambda = \Delta_n(\lambda)\langle n^{(0)}|n\rangle_\lambda \quad (\text{I})$$

we then obtain

$$|n\rangle_\lambda = C_n(\lambda)|n^{(0)}\rangle + \frac{1}{E_n^{(0)} - \hat{H}_0}\phi_n(\lambda \hat{V} - \Delta_n(\lambda))|n\rangle_\lambda \quad (\text{II})$$

and we have also obtained

$$C_n(\lambda) = \langle n^{(0)}|n\rangle_\lambda \quad (\text{III})$$

We now perform an expansion,

$$|n\rangle_\lambda = C_n(\lambda)\left(|n^{(0)}\rangle + \lambda|n^{(1)}\rangle + \lambda^2|n^{(2)}\rangle + \dots\right)$$

Applying equation (IV) to (I), we have

$$\langle n^{(0)}|\lambda \hat{V}|n^{(0)}\rangle + \lambda \langle n^{(0)}|n^{(1)}\rangle + \lambda^2 \langle n^{(0)}|n^{(2)}\rangle + \dots = \Delta_n(\lambda) = \Delta_n^{(1)}\lambda + \Delta_n^{(2)}\lambda^2$$

Resultantly,

$$\boxed{\Delta_n^{(1)}\lambda = \langle n^{(0)}|\lambda \hat{V}|n^{(0)}\rangle}$$

Similary, wanting $\lambda^2 \Delta_n^{(2)}$, we need to know $\lambda|n^{(1)}\rangle$. Equating (II) and (IV),

$$\lambda|n^{(1)}\rangle + \lambda^2|n^{(2)}\rangle + \dots = \frac{1}{E_n^{(0)} - H_0}\phi_n(\lambda \hat{V} - \Delta_n(\lambda))\left(|n^{(0)}\rangle + \lambda|n^{(1)}\rangle + \dots\right)$$

Leading us to

$$\boxed{\lambda|n^{(1)}\rangle = \frac{1}{E_n^{(0)} - H_0}\phi_n\lambda \hat{V}|n^{(0)}\rangle}$$

Applying to the above logic,

$$\langle n^{(0)}|\lambda \hat{V}|\lambda n^{(1)}\rangle = \Delta_n^{(2)}\lambda^2$$

and

$$\boxed{\Delta_n^{(2)}\lambda^2 = \langle n^{(0)}|\lambda \hat{V}\frac{1}{E_n^{(0)} - H_0}\phi_n\lambda \hat{V}|n^{(0)}\rangle}$$

Well, how can I express this?

$$\Delta_n^{(2)} \lambda^2 = \sum_{k \neq n} \frac{|\langle k^{(0)} | \lambda \hat{V} | n^{(0)} \rangle|^2}{E_n^{(0)} - E_k^{(0)}}$$

We cannot ignore $|n\rangle_\lambda$. We know that

$$\begin{aligned} {}_\lambda \langle n | n \rangle_\lambda = 1 &= |C_n(\lambda)|^2 \left(\langle n^{(0)} | + \lambda \langle n^{(1)} | + \lambda^2 \langle n^{(2)} | + \dots \right) \left(|n^{(0)}\rangle + \lambda |n^{(1)}\rangle + \lambda^2 |n^{(2)}\rangle + \dots \right) \\ &= |C_n(\lambda)|^2 \left(1 + \lambda^2 \langle n^{(1)} | n^{(1)} \rangle + \dots \right) \end{aligned}$$

and

$$|C_n(\lambda)|^2 = \frac{1}{1 + \lambda^2 \langle n^{(1)} | n^{(1)} \rangle + \dots} = 1 - \lambda^2 \langle n^{(1)} | n^{(1)} \rangle$$

Example. (Simple harmonic oscillator) Consider the solved harmonic oscillator with basis vectors $|n^{(0)}\rangle$ and $E^{(0)} = \hbar\omega(n + 1/2)$. Applying a uniform electric field, $E\hat{x}$,

$$\lambda \hat{V} = (-e)(-Ex) = eEx$$

For the first order, we have

$$\Delta_n^{(1)} = \langle n | \lambda \hat{V} | n \rangle = \langle n | eE\hat{x} | n \rangle = 0$$

For the second order, we have

$$\Delta_n^{(2)} = \sum_{k \neq n} \frac{|\langle k^{(0)} | eE\hat{x} | n^{(0)} \rangle|^2}{E_n - E_k} = -\frac{(eE)^2}{2m\omega^2} < 0$$

We know that this result, however, is exact. This is because of the form of the Hamiltonian,

$$H_0 + \lambda V = \frac{p^2}{2m} + \frac{1}{2}m\omega^2 x^2 + eEx = \frac{p^2}{2m} + \frac{m\omega^2}{2}(x - K)^2 - \frac{(eE)^2}{2m\omega^2}$$

where K is a constant. This is known as the Stark effect. Note that in the ground state,

$$\Delta_n^{(2)} = \sum_{k \neq n} \frac{|\langle k | \lambda \hat{V} | n \rangle|^2}{E_n - E_k} < 0$$

LECTURE 5 (SEPTEMBER 15TH)

Definition. (Degenerate perturbation theory) Notice that multiple states occupy the same energy level for the hydrogen atom with a Coulomb potential (e^2/r). Contrarily from the previous case where we saw one ket per energy level, we will try to see what happens when we deal with Hilbert spaces with multiple states per energy level. The state space with

identical energy levels can be expressed as (via projection)

$$P_0 = \sum_D |l_0\rangle\langle l_0|$$

where D is a degenerate subspace. The identity can thus be written as

$$\mathbf{1} = P_0 + P_1$$

Obviously,

$$\hat{H}_0 P_0 = \hat{H}_0 \sum_l |l_0\rangle\langle l_0| = E_D^{(0)} P_0$$

and

$$P_0 \hat{H}_0 = \sum_l |l_0\rangle\langle l_0| \hat{H}_0 = E_D^{(0)} P_0$$

and the two operators commute. We ultimately want to solve

$$\hat{H}|l\rangle = (\hat{H}_0 + \lambda \hat{V})|l\rangle = E|l\rangle$$

we state that it can be expressed as

$$|l\rangle = P_0|l\rangle + P_1|l\rangle$$

Applying the Hamiltonian, we expect

$$(\hat{H}_0 + \lambda \hat{V})(P_0|l\rangle + P_1|l\rangle) = E(P_0|l\rangle) + E(P_1|l\rangle)$$

Applying once more the projection operator P_0 ,

$$(E_D^{(0)} + P_0 \lambda \hat{V} P_0)P_0|l\rangle + P_0 \lambda \hat{V} P_1|l\rangle = E P_0|l\rangle$$

we can of course subtract the most-left term and arrive at

$$(P_0 \lambda \hat{V} P_0)P_0|l\rangle + P_0 \lambda \hat{V} P_1|l\rangle = (E - E_D^{(0)})P_0|l\rangle$$

where the energy difference on the right is called the energy spitting δE_0 . Ignoring the second term as we want only the 1st order, we have

$$(P_0 \lambda \hat{V} P_0)P_0|l\rangle = \delta E_0 P_0|l\rangle$$

We can also apply P_1 from the left to prove that indeed the second term is of second order and beyond. Having

$$\begin{aligned} P_1 \lambda \hat{V} P_0|l\rangle + (\hat{H}_0 + P_1 \lambda \hat{V} P_1)P_1|l\rangle &= E P_1|l\rangle \\ (E - \hat{H}_0 - P_1 \lambda \hat{V} P_1)P_1|l\rangle &= P_1 \lambda \hat{V} P_0|l\rangle \end{aligned}$$

which leads to

$$P_1|l\rangle = \frac{1}{E - H_0} P_1 \lambda V P_0 |l\rangle$$

Theorem. The following

$$\boxed{(P_0 \lambda V P_0) P_0 |l\rangle = \delta E_0 P_0 |l\rangle}$$

is ultimately the eigenvalue problem we want to solve to find the perturbed eigenkets. The energy is resultantly

$$\boxed{E = E_0^{(0)} + \delta E_0 + \lambda \sum_{k \notin D} \frac{|V_{kl}|^2}{E_D^{(0)} - E_k^{(0)}}$$

Example. (Isotropic rigid rotator) Consider the Hamiltonian

$$\hat{H}_0 = \frac{\vec{L} \cdot \vec{L}}{2I}$$

We know that the eigenkets and their energy levels are given as

$$|l, m\rangle \quad \& \quad \frac{l(l+1)\hbar^2}{2I}$$

If $l \neq 0$, degenerate $2l+1$. The perturbation will be given by

$$\hat{V} = -\vec{\mu} \cdot \vec{B} = -\frac{gqB}{2m} L_z$$

where $\vec{B}$ is an external magnetic field and $\vec{\mu} = gq\vec{L}/2m$. The g is dictated by whether its electron spin ($=2$) or whether its orbitals ($=1$). These kind of potentials are from Zeeman coupling. Now, we see that the potential is also commutative with L_z , as $[V, L_z] = 0$. We should utilise this! For a general state vector,

$$\langle m|[V, L_z]|m'\rangle = \langle m|VL_z - L_zV|m'\rangle = 0$$

which implies that

$$(m' - m)\hbar\langle m|V|m'\rangle = 0$$

Thus we have found the matrix element for the perturbation! We thus find

$$\frac{l(l+1)\hbar^2}{2I} - x\hbar m$$

for $x > 0$.

Definition. (Stark effect of the hydrogen atom) Assume no spin. For an Hydrogen atom in an electric field in the z -axis,

$$H_1 = (-e)V(r) = (-e)(-Ez) = eEz$$